Quantification: 701_z19c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	18
interior lacuna count	18
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 18)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1346.64	1045.5	887.752	px^2
major_axis_length_px	lacuna shape	62.9138	61.5882	22.8163	px
minor_axis_length_px	lacuna shape	30.9434	27.1289	12.7049	px
aspect_ratio	lacuna shape	2.1569	2.1545	0.7419	unitless
eccentricity	lacuna shape	0.81	0.8858	0.1882	unitless
solidity	lacuna shape	0.7688	0.743	0.0888	unitless
roots_count	headline	8.5	7.5	4.4225	count
ring_length_r30_px	headline	332.833	298.5	142.855	px
ring_length_r60_px	reported	743.833	705.5	343.168	px
owned_length_px	ownership-dependent	1023.22	639.916	904.08	px
edge_count	ownership-dependent	64.3333	37	57.3985	count
mean_edge_length_px	ownership-dependent	15.9629	15.729	2.5848	px
perimeter_px	normalised	205.908	184.501	111.612	px
ring_area_r30_px2	normalised	6719.39	6611	2126.84	px^2
ring_area_r60_px2	normalised	15468.4	15556	5475.23	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9954	1	0.0162	unitless
ring_density_r30	normalised	0.0488289	0.0478108	0.00979242	px^-1
ring_density_r60	normalised	0.0470249	0.0459008	0.0100821	px^-1
roots_per_100px_perimeter	normalised	4.4485	4.1462	1.8312	per 100 px
ring_attached_length_r30_px	network v2	288.556	239.5	148.498	px
ring_attached_length_r60_px	network v2	536.056	425.5	343.08	px
ring_length_w_r30_px	network v2	373.985	343.049	158.883	px
ring_length_w_r60_px	network v2	835.285	793.29	384.967	px
sholl_crossings_r10	network v2	8.5556	7	3.9291	count
sholl_crossings_r20	network v2	8.6667	7	3.6622	count
sholl_crossings_r30	network v2	9.0556	9.5	4.0652	count
loop_count	loops, report only	3.6667	1	5.688	count
loop_edge_share	loops, report only	0.1307	0.1265	0.143	share
ring_width_mean_r30_px	width	5.8752	6.09	1.638	px

Per field
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measure	value	units
analysed_area_px2	796022	px^2
total_skeleton_length_px	34723	px
canalicular_length_density_per_px	0.0436206	px^-1
junction_count	912	count
junction_density_per_px2	0.0011457	px^-2
lacuna_count	18	count

lacunae_per_px2	2.261e-05	px^-2
skeleton_component_count	591	count
mean_component_length_px	58.753	px
median_component_length_px	12	px
field_length_density_w_per_px	0.049009	px^-1
field_density_without_flagged_per_px	0.0282018	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	394	px
unattached_component_count	21	count
canalicular_length_density_attached_per_px	0.0436206	px^-1
field_length_density_with_unattached_per_px	0.0441156	px^-1
excluded_area_px2	232055	px^2
analysed_field_area_px2	816521	px^2
loop_count	125	count
loop_edge_share	0.1924	share
edge_count	2994	count

Canalicular width, the new measure

measure	value	units
width_mean_px	6.6128	px
width_median_px	6.3246	px
width_p10_px	2.8284	px
width_p90_px	10.198	px
width_pixels_used	22195	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	653	48.9306	18.0087	2.7171	0.9298	0.8718	1.3396	244.608	191.13	False
2	439	24.8605	23.4655	1.0594	0.3303	0.8905	-1.2149	286.371	223.007	False
3	511	42.2275	16.9827	2.4865	0.9156	0.8705	0.9114	308.108	170.11	False
4	663	44.6399	20.5821	2.1689	0.8874	0.8511	1.5097	320.038	647.3	False
5	1055	56.5297	24.2615	2.33	0.9032	0.8843	1.0383	419.101	230.957	False
6	1893	74.136	39.3577	1.8836	0.8474	0.6949	-1.3345	688.738	268.372	False
7	1441	68.9456	33.0049	2.089	0.878	0.7238	-1.5587	747.817	126.852	False
8	3491	77.9644	65.3058	1.1938	0.5462	0.7932	-1.5218	773.022	646.082	False
9	988	66.6467	25.2664	2.6378	0.9254	0.6694	-1.3153	754.104	873.76	False
10	1270	71.948	33.6183	2.1401	0.8841	0.6499	-1.4065	891.32	930.428	False
11	1577	97.9	24.8484	3.9399	0.9673	0.7152	-1.2598	906.939	130.226	False
12	957	43.0583	35.4718	1.2139	0.5669	0.6965	1.5597	917.457	535.305	False
13	1036	53.8662	28.9914	1.858	0.8428	0.7623	-1.4483	935.929	453.511	False
14	2879	109.14	44.0418	2.4781	0.915	0.6894	-1.4893	959.511	178.398	False
15								361.992	272.702	False
16								528.416	314.222	False
17								549.191	341.457	False
18								659.583	359.618	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	4	206	443	371.22	20	18.561	3.2714
2	6	195	253	206.61	19	10.8742	5.9249
3	3	201	350	325.978	16	20.3736	5.9292
4	5	239	618	454.505	36	12.6251	6.6219
5	7	270	573	667.328	38	17.5613	5.7057
6	9	345	832	1526.94	107	14.2705	6.6405
7	8	403	1011	842.914	48	17.5607	6.6716
8	15	545	1425	3114.19	197	15.8081	7.4235
9	8	327	793	438.818	24	18.2841	6.1351
10	12	408	1089	1726.18	86	20.0718	6.0449
11	16	541	1106	1853.89	116	15.9818	7.8099
12	12	424	896	1459.04	95	15.3583	2.8284
13	11	401	995	2723.06	174	15.6498	4.5701
14	17	658	1221	1558.89	108	14.4341	7.502
15	7	189	393	205.723	14	14.6945	2.4142
16	3	222	443	222.61	18	12.3672	6.4028
17	4	205	447	107.498	7	15.3568	5.7734
18	6	212	501	612.505	35	17.5001	8.0847

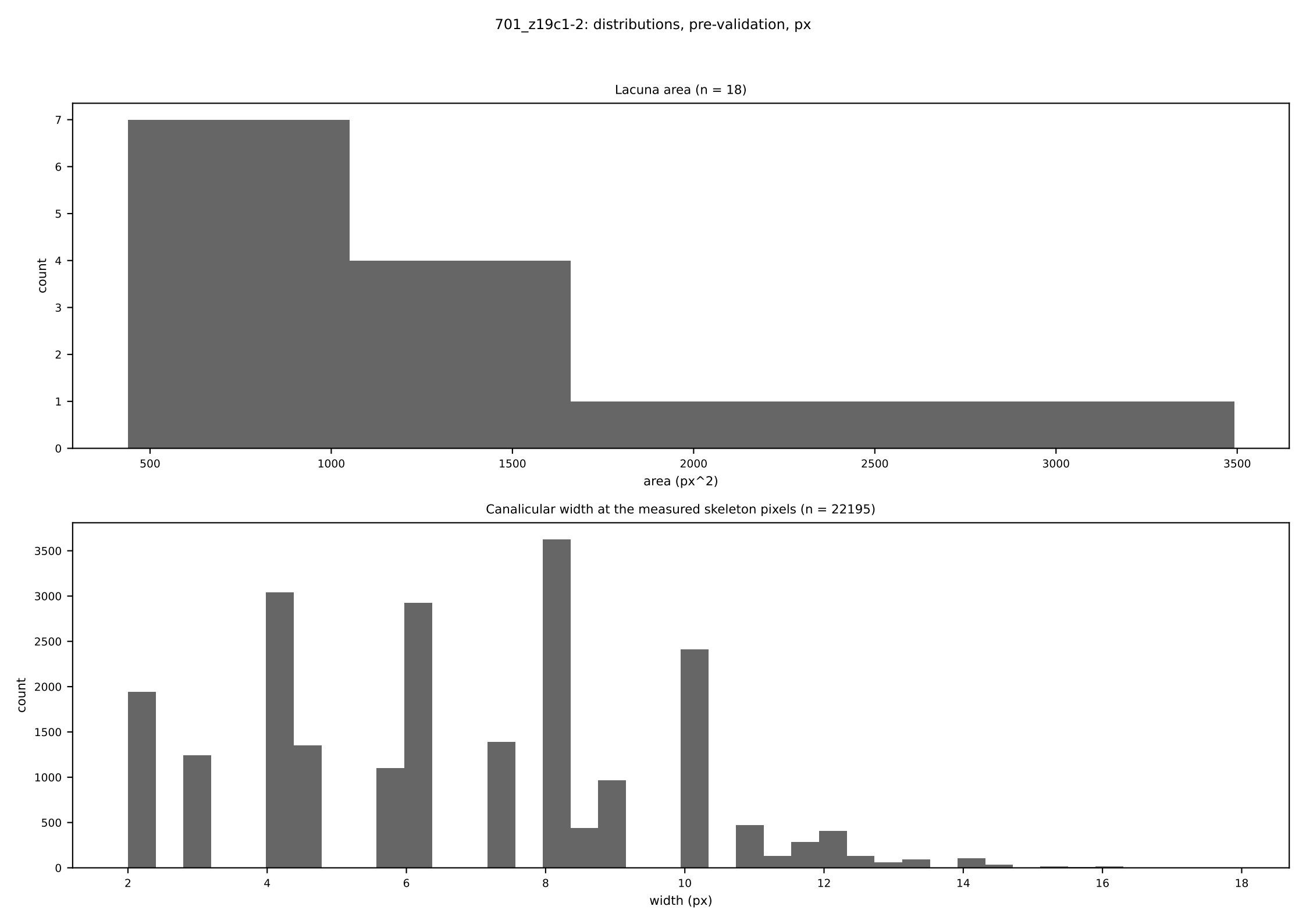
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	122.225	4947	11212	1	1	0.0416414	0.0395112	3.2726
2	94.7107	3651	5943	1	1	0.05341	0.0425711	6.3351
3	114.711	4934	9822	1	1	0.0407377	0.0356343	2.6153
4	134.61	6194	17999	1	1	0.0385857	0.0343352	3.7144
5	162.894	6766	14585	1	1	0.0399054	0.0392869	4.2973
6	260.777	8904	19018	1	1	0.0387466	0.043748	3.4512
7	308.818	8500	21039	1	1	0.0474118	0.0480536	2.5905
8	337.446	10732	26608	1	1	0.0507827	0.0535553	4.4452
9	200.25	7746	20901	1	1	0.0422153	0.0379408	3.995
10	267.984	8463	21768	1	0.9865	0.0482098	0.0500276	4.4779
11	301.404	8091	17268	1	1	0.0668644	0.0640491	5.3085
12	255.806	6547	15232	1	1	0.0647625	0.0588235	4.6911
13	199.622	6675	15880	1	1	0.0600749	0.0626574	5.5104
14	503.269	10225	19270	1	0.9316	0.0643521	0.0633627	3.3779
15	97.8823	5235	10032	1	1	0.0361032	0.0391746	7.1514
16	169.38	5166	12626	1	1	0.0429733	0.0350863	1.7712
17	111.196	3920	9020	1	1	0.0522959	0.0495565	3.5973
18	63.3553	4253	10209	1	1	0.0498472	0.0490744	9.4704

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	193	302	234.238	494.718	5	5	5
2	148	152	218.51	281.38	6	5	2
3	147	221	232.137	405.233	5	7	6
4	206	297	267.066	690.014	8	6	7
5	237	382	312.392	638.671	7	7	7
6	247	513	382.321	945.637	7	13	12
7	333	747	448.363	1126.75	10	10	13
8	509	1136	606.061	1588.99	13	15	14
9	242	404	373.706	896.566	6	12	9
10	327	714	456.433	1228.84	9	10	15
11	537	1066	602.576	1241.02	19	13	10
12	417	728	483.233	1015.44	9	12	10

13	367	923	451.262	1124.82	12	6	10
14	625	1083	737.357	1368.36	15	14	17
15	159	162	214.309	443.475	6	4	4
16	192	236	241.267	490.647	5	6	5
17	136	136	230.995	505.375	7	5	7
18	172	447	239.51	549.191	5	6	10


701_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 18)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22195)
count
width (px)